

Lab Program 5

Demonstrate CI integration by running jenkins in a docker container

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Step 1 : Verify if the required software exist

```
docker --version
docker compose --version
```

Step 2 : Create the required files

```
mkdir project
cd project
touch Dockerfile docker-compose.yml
ls
```

Step 3 Required folder structure

```
> pwd
/home/1RV24MC042_harshavardhan_b_r/Desktop/RVCE/devopslab/t2_p5
> tree
.
├── docker-compose.yml
└── Dockerfile

1 directory, 2 files
```

Step 4 : add content to the files

```
// docker-compose.yml
version: "3.8"

services:
  jenkins:
    container_name: program5-ctr
    build: .
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home

volumes:
  jenkins_home:
```

```
// Dockerfile
FROM jenkins/jenkins:lts

USER root

RUN apt-get update && apt-get install -y git

USER jenkins
```

Step 5 : Build run and verify

```
docker compose build
docker images

docker compose up -d
docker ps
docker compose down -v
```

```

> ls
docker-compose.yml  Dockerfile
> docker build -t p5 .
[+] Building 3.4s (3/3) FINISHED          docker:default
=> [internal] load build definition from Dockerfile  0.0s
=> => transferring dockerfile: 169B  0.0s
=> CANCELED [internal] load metadata for docker.io/j  3.4s
=> [auth] jenkins/jenkins:pull token for registry-1.  0.0s
ERROR: failed to build: failed to solve: Canceled: context c
anceled
>
>
> docker compose up -d
WARN[0000] /home/1RV24MC042_harshavardhan_b_r/Desktop/RVCE/d
evopslab/p5/docker-compose.yml: the attribute `version` is o
bsolete, it will be ignored, please remove it to avoid poten
tial confusion
[+] Running 3/3
 ✓ Network p5_default          C...          0.1s
 ✓ Volume p5_jenkins_home      Created          0.0s
 ✓ Container jenkins-server    Started          0.3s
> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED
              STATUS        PORTS
              NAMES
935b7f308c35   p5-jenkins    "/usr/bin/tini -- /u..." About a
minute ago    Up About a minute    0.0.0.0:8080->8080/tcp, [:
:]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->500
00/tcp        jenkins-server

```

For the password

```

> docker exec -it 935b /bin/bash
jenkins@935b7f308c35:/$ cat /var/jenkins_home/secrets/initialAdminPassword

e85ae715ccc414006af378d42bbad19d4
jenkins@935b7f308c35:/$

```